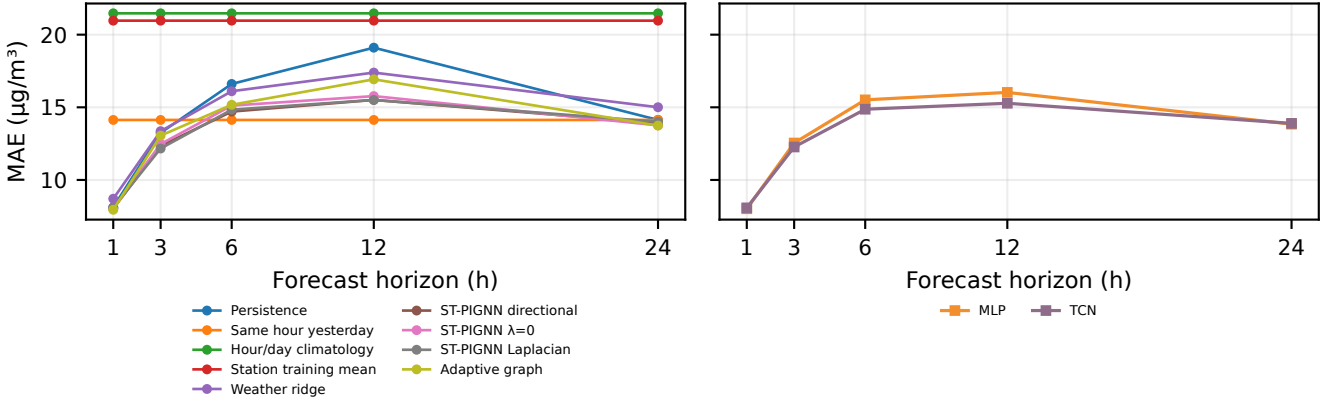
A Retrospective centered-weather protocol
B Forecast-cutoff-safe MLP/TCN protocol
MAE (µg/m³)
10
15
20
1
3
6
12
24
Forecast horizon (h)
Persistence
Same hour yesterday
Hour/day climatology
Station training mean
Weather ridge
ST-PIGNN directional
ST-PIGNN λ=0
ST-PIGNN Laplacian
Adaptive graph
MLP
TCN